HRM Pro Enterprise

Complete Application Quality Assurance & RBAC Audit Specification

QA Specification & Test Execution Plan
System Version: 4.2.0 Enterprise Release
Local App URL: `http://localhost:3000`
Target: Chrome Automation & Claude Testing

Complete Application RBAC & QA Test Suite

This master document provides comprehensive, end-to-end test cases covering the entire **HRM Pro** application across all user personas, permission tiers, business modules, and Pakistani statutory compliance workflows.

> **Local Environment & System Endpoints:**
> • **Primary Web App & SPA Frontend:** `http://localhost:3000`
> • **REST API Base:** `http://localhost:3000/api` | **Health Check:** `http://localhost:3000/api/health`
> • **Live WebSocket Stream:** `ws://localhost:3000/ws` | **Sync SSE Stream:** `http://localhost:3000/api/sync/stream`
> • **Backend Storage:** Autonomous Dual Storage (Express local store with Supabase PostgreSQL cloud sync fallback)

1. Seed User Personas & Credentials Matrix

Username	Password	Role / Persona	Employee Record	Entity / Dept Scope	Authority Tier & Privileges
`admin`	`admin123`	SUPERADMIN	EMP-001 (Ahmed Khan, CEO)	All Holdings (Global)	**Universal Master Access:** Sovereign access to all 16 modules, multi-company switcher, payroll finalization, administrative security.
`sara.malik`	`hr123`	HR_MANAGER	EMP-002 (Sara Malik, HR Head)	Apex Technologies (Entity)	**Corporate HR Authority:** Employee lifecycle, e-DMS records, ATS recruitment pipeline, Tier-2 final leave approvals, payroll runs.
`usman.baig`	`mgr123`	DEPT_MANAGER	EMP-003 (Usman Baig, Tech Lead)	IT & Engineering Dept	**Team Line Management Scope:** Scoped team telemetry, Tier-1 leave endorsements, missing punch regularization. Blocked from company salaries.
`fatima.raza`	`emp123`	EMPLOYEE	EMP-004 (Fatima Raza, Software Eng)	IT & Engineering Dept	**Strict Self-Service Only:** Biometric punch clock, apply personal leaves, view own confidential payslips with FBR tax. Peer data strictly isolated.
`saad.ibrahim`	`emp123`	ONBOARDING	EMP-026 (Saad Ibrahim, New Hire)	Corporate Induction	**New Joiner Onboarding Vault:** Upload induction documents (CNIC, degrees, IBAN), acknowledge handbook. Operational modules hidden.
`junior.hr`	`hr123`	HR_MANAGER (Restricted)	EMP-006 (Zain Ali, Junior HR)	HR Operations Desk	**Partial Permissions Desk:** Read employee directory, log candidate notes. High-privilege actions (salary finalize, system settings) revoked.

2. Navigation Architecture (Two-Tier Persistent Header)

> **Layout Verification Standard:**
> • **Tier 1 (Top Bar):** Brand Logo, Global Search (`Ctrl+K`), Multi-Company Switcher, Demo Persona Switcher, Dark/Light Mode, Notifications.
> • **Tier 2 (Main Category Pillars):** `DASHBOARD` | `PEOPLE & TALENT` | `TIME & ATTENDANCE` | `FINANCE & PAYROLL` | `OPERATIONS & ADMIN`
> • **Tier 3 (Persistent Sub-Navigation Bar):** Clicking any category pillar renders horizontal flat sub-tabs underneath. **Zero hover dropdowns obscuring the screen**.

3. Complete Application Test Suites by Module

Suite 1: Authentication, Session Management & Persona Switcher

Test ID	Persona	Action / Steps	Expected Result
AUTH-01	`admin`	Navigate to `http://localhost:3000/#login`. Enter `admin` / `admin123`.	Login succeeds. Redirects to `#dashboard`. Topbar displays Ahmed Khan (Superadmin). Session stored in localStorage.
AUTH-02	`sara.malik`	Login with `sara.malik` / `hr123`.	Lands on Dashboard with HR Manager role badge. Corporate HR widgets loaded.
AUTH-03	`usman.baig`	Login with `usman.baig` / `mgr123`.	Dashboard scopes strictly to IT & Engineering team members and pending manager approvals.
AUTH-04	`fatima.raza`	Login with `fatima.raza` / `emp123`.	Dashboard renders Self-Service view: Punch Clock, personal leave balance, own upcoming events.
AUTH-05	Any Persona	Enter valid username with incorrect password (e.g. `admin` / `wrongPass`).	Login blocked with error: "Invalid username or password". Password field cleared.
AUTH-06	Inactive User	Attempt login with deactivated user (`inactive.user` / `Password123!`).	Rejection alert: "Account Inactive: Your employee profile has been deactivated."
AUTH-07	Any Persona	Click User Avatar → **Sign Out**.	Session purged from sessionStorage and localStorage. Immediate redirect to `#login`.
AUTH-08	Any Persona	Select persona from **Demo Persona Switcher** in topbar.	Instant session swap to chosen persona without page reload; navigation updates to match role.

Suite 2: Executive Dashboard & Live Biometric Punch Clock

Test ID	Persona	Action / Steps	Expected Result
DASH-01	`admin`	Inspect 4 KPI Summary Cards on Dashboard.	Total Employees, Present Today, On Leave, Monthly Payroll show live numbers with percentage sparklines. Zero `NaN` or `undefined`.
DASH-02	`fatima.raza`	Inspect Live Punch Clock widget. Click **Clock In**.	Live seconds timer running. Click plays confirmation audio chime, logs punch timestamp, and button toggles to glowing **Clock Out**.
DASH-03	`fatima.raza`	Click **Clock Out**.	Calculates shift working duration, logs out-time, and shows success toast.
DASH-04	`admin`	Inspect **Executive Action Inbox** widget.	Displays items awaiting sign-off (pending leaves, expense claims, new hires). Badge counts reflect live pending items.
DASH-05	Any	Inspect **Corporate Headlines & Noticeboard** widget.	Displays current announcements, upcoming holidays, and company news cleanly.

Test ID	Persona	Action / Steps	Expected Result
DASH-06	Any	Click **Floating WhatsApp / Quick Action** button.	Opens quick support dialogue or initiates WhatsApp helpline without layout breakage.

Suite 3: Workforce Directory & e-DMS Digital File Vault

Test ID	Persona	Action / Steps	Expected Result
EMP-01	`sara.malik` (HR)	Click `PEOPLE & TALENT` → `Employees & e-DMS`.	Full workforce directory displays with Avatar, Emp ID, Full Name, Designation, Department, Status, Actions.
EMP-02	`sara.malik` (HR)	Click **+ Add Employee**. Fill details and save.	Form validates inputs. New employee is added to database and audit log registers creation.
EMP-03	`sara.malik` (HR)	Click on `EMP-004 Fatima Raza` to open e-DMS Vault.	Displays digital file: Basic Info, Employment, Statutory (CNIC, NTN, EOBI), Bank (IBAN), and Document Attachments.
EMP-04	`usman.baig` (Mgr)	Click `PEOPLE & TALENT` → `Employees & e-DMS`.	Filtered strictly to IT & Engineering department. Employees in Finance, HR, or Marketing are hidden.
EMP-05	`fatima.raza` (Emp)	Click `PEOPLE & TALENT` → `My Profile`.	Only Fatima's own profile opens. Cannot view peer salaries or confidential records.
EMP-06	`junior.hr`	Inspect employee record actions.	View and note logging permitted; destructive actions (Delete, Terminate, Salary Edit) disabled.

Suite 4: Time, Attendance, Biometric Gateway & Muster Roll

Test ID	Persona	Action / Steps	Expected Result
ATT-01	`admin` / `sara.malik`	Click `TIME & ATTENDANCE` → `Attendance & Shifts`.	Daily attendance grid renders check-in, check-out, working hours, and punctuality badges (On Time, Late, Absent).
ATT-02	Any	Open **Biometric Simulator** tab. Click **Simulate Punch**.	Simulates TCP/IP ZKTeco machine punch. Pushes punch via WebSocket and updates attendance grid in real time.
ATT-03	`fatima.raza` (Emp)	Click **Request Attendance Regularization**.	Submits missed punch regularisation request with reason. Status set to Pending Manager Approval.
ATT-04	`usman.baig` (Mgr)	Open Regularization Approvals queue.	Usman sees Fatima's request. Can click **Endorse** or **Reject** with manager comments.
ATT-05	`admin` / `sara.malik`	Open **Monthly Statutory Muster Roll** tab.	Generates 31-day attendance matrix with P/A/L/H codes, total payable days, and CSV export.
ATT-06	Any	Toggle GPS Geofencing in Attendance Settings.	Punches recorded outside office geofence radius are flagged as "Remote / Geofence Warning".
ATT-07	`fatima.raza` (Emp)	Simulate Offline Mode in DevTools. Click Clock In.	PWA service worker queues punch locally; automatically syncs to backend upon network reconnection.

Suite 5: Leave Management & 2-Tier Approval Workflow

Test ID	Persona	Action / Steps	Expected Result
LEV-01	`fatima.raza` (Emp)	Click `TIME & ATTENDANCE` → `Leaves & Absence`.	Quota cards display Annual (14), Casual (10), Sick (8) with Used, Available, and Pending numbers.

Test ID	Persona	Action / Steps	Expected Result
LEV-02	`fatima.raza` (Emp)	Click **Apply Leave**. Select Casual Leave (2 days), enter reason, and Submit.	Leave application recorded. Status set to `Pending Manager Endorsement`.
LEV-03	`usman.baig` (Mgr)	Log in as `usman.baig`. Open Pending Leave Approvals.	Fatima's request appears in queue. Usman clicks **Endorse Request**. Status advances to `Pending HR Approval`.
LEV-04	`sara.malik` (HR)	Log in as `sara.malik`. Open Executive Leave Approvals.	Request displays with Manager Endorsement. Sara clicks **Final Approve**. Balance deducted; status `Approved`.
LEV-05	`usman.baig` (Mgr)	Inspect leave queue for non-IT employees (Finance, HR).	Non-IT employee leaves do NOT appear in Usman's queue. Department isolation verified.
LEV-06	`admin` / `sara.malik`	Inspect **Team Leave Heatmap & Calendar**.	Visual calendar highlights planned leaves, preventing cross-team coverage bottlenecks.

Suite 6: Finance, Payroll & Pakistani Statutory FBR Tax Engine

Test ID	Persona	Action / Steps	Expected Result
PAY-01	`admin` / `sara.malik`	Click `FINANCE & PAYROLL` → `Payroll & Taxes` → **Generate Payroll Run**.	Calculates Basic Pay, House Rent (45%), Medical (10%), Utilities (10%), Gross Salary for all active employees.
PAY-02	`admin`	Verify FBR Tax calculation for PKR 150,000 monthly salary (PKR 1.8M annual).	FBR 2025-27 progressive salaried tax table computes dynamic monthly withholding. Slabs above 600K apply incrementally.
PAY-03	`admin`	Inspect statutory deductions: Provident Fund (PF 8.33%) and EOBI (PKR 350).	Net Salary matches: `Gross - FBR Tax - PF - EOBI - Unpaid Days`. Deductions ledger balanced.
PAY-04	`admin` / `sara.malik`	Click **Generate Bank Disbursal File**. Select HBL / MCB / 1Link format.	Exports formatted CSV/Excel file with Account No, IBAN, Beneficiary Name, Net Salary, and batch checksum.
PAY-05	`fatima.raza` (Emp)	Log in as `fatima.raza`. Navigate to `FINANCE & PAYROLL`.	Fatima sees ONLY her personal payslips. **All other employee salaries are completely hidden**.
PAY-06	`fatima.raza` (Emp)	Click **Download PDF Payslip**.	Generates branded PDF payslip with salary breakdown, tax deductions, and net disbursed figure.
PAY-07	`usman.baig` (Mgr)	Navigate to `FINANCE & PAYROLL`.	Cannot view salaries of peers or department heads. Payroll execution buttons hidden.

Suite 7: Statutory Gratuity Final Settlement (30/26 Formula)

Test ID	Persona	Action / Steps	Expected Result
SET-01	`admin` / `sara.malik`	Click `FINANCE & PAYROLL` → `Final Settlement` → **Initiate Exit Clearance**.	Settlement wizard opens: Select employee, Exit Date, Notice Period Served, and Reason for Exit.
SET-02	`admin` / `sara.malik`	Inspect **Statutory Gratuity Card**.	Computes Pakistani labor law formula: `(Last Gross Salary ÷ 26 × 30) × Qualifying Service Years` accurately.
SET-03	`admin` / `sara.malik`	Verify **3-Gate Clearance Workflow** (Gate 1: IT Asset, Gate 2: Admin, Gate 3: Finance).	Each gate requires sign-off with remarks. Final voucher blocked until all 3 gates are marked cleared.
SET-04	`admin` / `sara.malik`	Inspect Leave Encashment & Notice Period Adjustments.	Unavailed leaves encashed at last gross daily rate; unserved notice period deducted proportionately.

Test ID	Persona	Action / Steps	Expected Result
SET-05	`admin`	Click **Generate Settlement Voucher (Print/PDF)**.	Outputs clearance letter and settlement voucher with itemized gratuity, deductions, and signature blocks.

Suite 8: Multi-Company Model A Scoping (Holding & Subsidiaries)

Test ID	Persona	Action / Steps	Expected Result
CMP-01	`admin`	In Topbar Company Switcher, switch from `Apex Group` to `Apex FinTech Ltd`.	UI scopes instantly to Apex FinTech. Employee directory and dashboard numbers reload to reflect only this subsidiary.
CMP-02	`admin`	Click `OPERATIONS & ADMIN` → `Company Entities`.	Superadmin can manage holding companies, add branches, and assign entity HR managers.
CMP-03	`sara.malik` (HR)	Inspect Topbar Company Switcher.	Sara is locked to her assigned company (`Apex Technologies`). Sister entity records remain isolated.
CMP-04	`admin`	Switch back to `Apex Group (All Holdings)`.	Aggregates group-level headcount and global payroll metrics.

Suite 9: Recruitment ATS & Kanban Pipeline

Test ID	Persona	Action / Steps	Expected Result
REC-01	`sara.malik` (HR)	Click `PEOPLE & TALENT` → `Recruitment (ATS)`.	Kanban board displays columns: Screening, Shortlisted, Technical Interview, HR Interview, Job Offer, Hired.
REC-02	`sara.malik` (HR)	Click **+ Post Job Opening**. Fill title, department, vacancies and publish.	Job requisition created and visible in recruitment dashboard.
REC-03	`sara.malik` (HR)	Drag candidate card from Screening to Technical Interview.	Candidate card moves smoothly. Prompt opens to schedule interview with Department Manager.
REC-04	`sara.malik` (HR)	Click **Generate Offer Letter** on candidate in Job Offer column.	Generates branded offer letter with proposed designation, compensation, and joining date.
REC-05	`sara.malik` (HR)	Click **Mark Hired & Convert to Employee**.	Provisions new employee record into e-DMS directory and generates onboarding checklist.

Suite 10: Performance Management, OKRs & Appraisals

Test ID	Persona	Action / Steps	Expected Result
PRF-01	`usman.baig` (Mgr)	Click `PEOPLE & TALENT` → `Performance & OKRs`.	Usman views assigned KPIs and sprint goals for his IT engineering team members.
PRF-02	`usman.baig` (Mgr)	Conduct manager appraisal review for Fatima Raza. Enter ratings (1-5) and submit.	Review saved. Weighted score computed; status updates to Completed.
PRF-03	`fatima.raza` (Emp)	Log in as `fatima.raza`. Navigate to Performance & OKRs.	Fatima views her personal goals and manager appraisal notes. Peer ratings remain hidden.
PRF-04	`admin` / `sara.malik`	Open Annual Promotion & Increment Simulator.	Simulates performance-linked salary revisions and generates increment letters.

Suite 11: Company Assets & Inventory Lifecycle

Test ID	Persona	Action / Steps	Expected Result
AST-01	`admin` / `sara.malik`	Click `TIME & ATTENDANCE` → `Assets & Inventory`.	Inventory list displays assets (Laptops, Chairs, Phones) with Asset Tag, Serial, Status (Available/Assigned).
AST-02	`sara.malik` (HR)	Assign asset `ASSET-001 (Dell Laptop)` to `EMP-004 Fatima Raza`.	Status updates to Assigned; assignment timestamp and handover agreement logged.
AST-03	`fatima.raza` (Emp)	Log in as Fatima. Check My Profile → Assigned Assets.	Dell Laptop appears in Fatima's personal custody list.
AST-04	`sara.malik` (HR)	Initiate asset return.	Asset returned to inventory; condition logged; updates exit settlement clearance if retiring.

Suite 12: Expense Claims & Travel Reimbursements

Test ID	Persona	Action / Steps	Expected Result
EXP-01	`fatima.raza` (Emp)	Click `FINANCE & PAYROLL` → `Expense Claims` → **Submit Claim**.	Enter Category (Travel/Fuel), Amount (PKR 4,500), attach receipt, and Submit.
EXP-02	`fatima.raza` (Emp)	Inspect personal claims table.	Claim appears with status Pending Manager Endorsement.
EXP-03	`usman.baig` (Mgr)	Log in as `usman.baig`. Open Team Expense Approvals.	Fatima's claim displays with receipt preview. Usman clicks Endorse Claim.
EXP-04	`admin` / `sara.malik`	Approve claim in Finance queue.	Claim approved; reimbursement amount automatically queues into upcoming monthly payroll slip.

Suite 13: Helpdesk Ticketing, Live Chat & Gemini HR Co-Pilot

Test ID	Persona	Action / Steps	Expected Result
HLP-01	`fatima.raza` (Emp)	Click `OPERATIONS & ADMIN` → `Helpdesk & Support` → **Create Ticket**.	Submit ticket (Category: IT Support / VPN, Priority: High). Ticket registered with tracking ID (e.g. TICK-101).
HLP-02	`admin` / `usman.baig`	Open Helpdesk dashboard. Assign ticket to technician and mark Resolved.	Ticket status updates live; employee receives resolution notification.
HLP-03	Any Persona	Open floating **Enterprise Live Chat** on bottom-right. Send message.	Transmits via WebSocket, timestamps message, updates chat thread with zero latency.
HLP-04	Any Persona	Open **HR AI Co-Pilot (Gemini Assistant)**. Ask about gratuity formula.	Gemini model (gemini-3.5-flash-lite or fast fallback) explains (Last Gross/26*30)*Years clearly with no errors.
HLP-05	Any Persona	Click floating **WhatsApp Quick Connect** launcher.	Opens pre-configured WhatsApp chat link with HR helpline.

Suite 14: Headcount Analytics & Custom Reports

Test ID	Persona	Action / Steps	Expected Result
REP-01	`admin` / `sara.malik`	Click `OPERATIONS & ADMIN` → `Reports & Analytics`.	Dashboard renders Chart.js visualisations: Department Headcount, Diversity, Attrition, Attendance trends.
REP-02	`admin` / `sara.malik`	Generate **FBR Annual Tax Schedule Report**.	Generates annual tax ledger with employee CNICs, gross taxable salaries, and progressive tax deducted.

Test ID	Persona	Action / Steps	Expected Result
REP-03	`admin` / `sara.malik`	Select **Attendance Muster Summary** and export to CSV.	Downloads clean CSV file with zero encoding errors.
REP-04	`usman.baig` (Mgr)	Navigate to Reports & Analytics.	Usman can only generate reports pertaining to his own team members (Team Attendance, Team Leaves).

Suite 15: Administration, RBAC Role Matrix & Security Audit Trail

Test ID	Persona	Action / Steps	Expected Result
ADM-01	`admin` (Superadmin)	Click `OPERATIONS & ADMIN` → `Administration`.	Access allowed: User Accounts, Role & Permissions Matrix, Departments, Branches, Designations, Audit Logs.
ADM-02	`admin` (Superadmin)	In User Accounts tab, click **Reset Password** on a demo user.	Password resets securely; audit log registers action with admin timestamp and IP.
ADM-03	`admin` (Superadmin)	Inspect **Role & Permissions Matrix**.	Toggles accurately reflect permission rules (View, Create, Edit, Delete, Approve) across modules.
ADM-04	`admin` (Superadmin)	Inspect **Security Audit Logs** tab.	Displays immutable audit trail with User, Action, Entity, Details, and Timestamp.
ADM-05	`sara.malik` / `usman.baig`	Attempt to navigate to `#administration`.	Access restricted. System administration settings reserved exclusively for Superadmin.

Suite 16: System Settings, Themes, Density & PWA Offline Queue

Test ID	Persona	Action / Steps	Expected Result
SET-01	Any Persona	Toggle between **Obsidian Dark Mode** and **Crisp Light Mode**.	Instant theme transition with high-contrast readability; persists across page reload in localStorage.
SET-02	Any Persona	Toggle **Table Density** (High-Density Compact vs Comfortable).	Tables immediately toggle row padding without page refresh, optimising accounting data view.
SET-03	Any Persona	Press `Ctrl+K` (or `Cmd+K`).	Spotlight Command Palette opens. Typing module name navigates directly. Chords G+D, G+E, G+P work.
SET-04	Any Persona	Click **Interactive Guided Tour**.	Interactive walkthrough highlights navigation pillars, punch clock, action inbox, and profile.
SET-05	Any Persona	Inspect Service Worker registration (`sw.js`).	PWA service worker active, enabling offline punch queue and static asset caching.

4. Security & Boundary Negative Tests

Test ID	Persona	Unauthorized Attempt	Expected Secure Behavior
SEC-01	`fatima.raza` (Emp)	Direct URL jump: typing `http://localhost:3000/#administration`.	Blocked: System intercepts navigation, shows access warning toast, redirects to safe view.
SEC-02	`fatima.raza` (Emp)	Inspect Console: attempting to read `DB.get('users')` or peer salaries.	Passwords and hashes sanitized from memory. Cross-employee salary records are inaccessible.
SEC-03	`usman.baig` (Mgr)	Attempt to endorse a leave outside IT department.	Blocked: Non-department requests do not appear in manager inbox.
SEC-04	`saad.ibrahim` (Onboarding)	Attempt to access Payroll, ATS, or Attendance modules.	Blocked: Onboarding persona locked strictly to induction vault and handbook acknowledgment.

Test ID	Persona	Unauthorized Attempt	Expected Secure Behavior
SEC-05	Any Persona	Rapid clicking across navigation pillars in Row 2.	Instant sub-nav row update; zero dropdown overlapping or glitches.
SEC-06	Any Persona	XSS attempt in leave remarks (`<script>alert('xss')</script>`).	Sanitized via `Utils.escapeHtml()` before rendering. Zero script execution.
SEC-07	Inactive User	Attempting API requests with deactivated session.	API rejects with 401/403 and purges client session tokens.
SEC-08	Any Persona	Browser zoom testing (100%, 125%, 150%) and mobile viewport (375px).	Layout reflows seamlessly: Topbar collapses to mobile hamburger, bottom nav dock activates.

5. Chrome Claude Prompt Instructions

```
Please open http://localhost:3000. Follow the test plan in this document.
Test each persona sequentially:
1. admin (admin123) - Superadmin Universal Access, Multi-Company Switcher, Two-Tier Nav, FBR Tax, Gratuity
2. sara.malik (hr123) - HR Director, e-DMS Vault, ATS Kanban, Final Leave Approval
3. usman.baig (mgr123) - Line Manager, Scoped Team Telemetry, Gate-1 Approvals, No Company Salaries
4. fatima.raza (emp123) - Self-Service, Punch Clock with chime, Leave Submission, Own Payslip Only
5. saad.ibrahim (emp123) - Onboarding Induction Vault & Policy Acknowledgment
6. junior.hr (hr123) - Restricted Junior HR with partial permissions (Read/Log only)
Verify that all navigation uses the persistent two-tier bar without dropdowns, and report pass/fail status.
```